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(54) **CARD READER, SMART CARD AND
METHOD FOR PROCESSING A
TRANSACTION**

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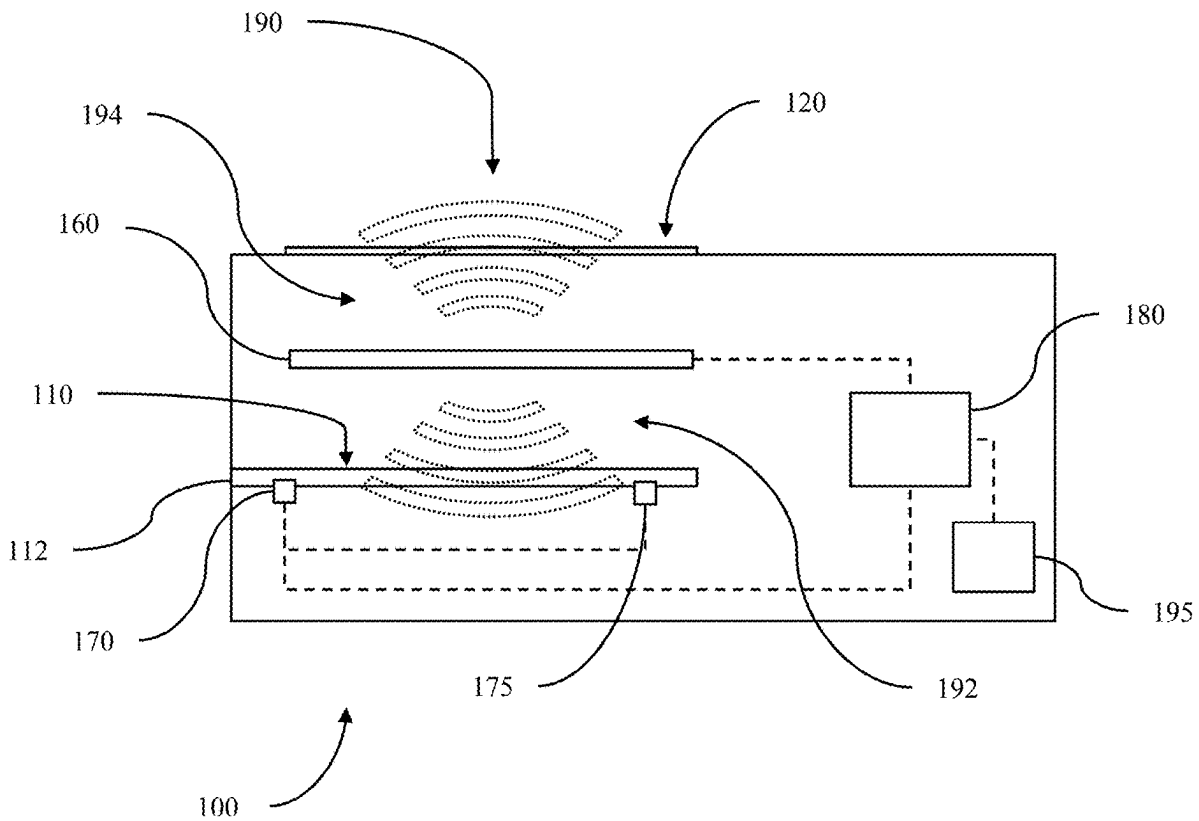
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(57) **ABSTRACT**

A card reader for reading a smart card, including an antenna for making a wireless connection with a smart card. The card reader further includes a card detector, for determining a location of the smart card relative to the antenna. The card reader has a processor configured to provide communication with the smart card and to allow or provide a first wireless communication when the card is determined to be at a first location of the card reader and to provide a second wireless communication when the card is determined to be at a second, different, location of the card reader.



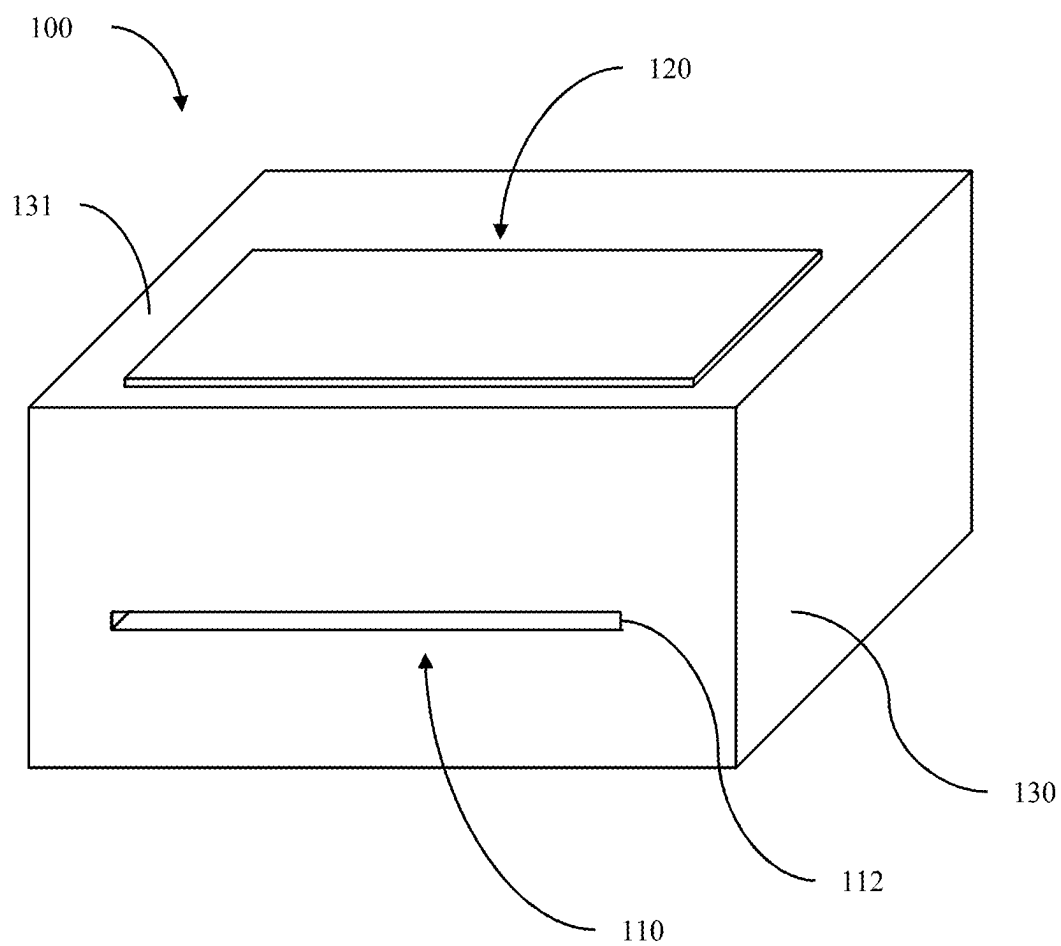


FIG. 1

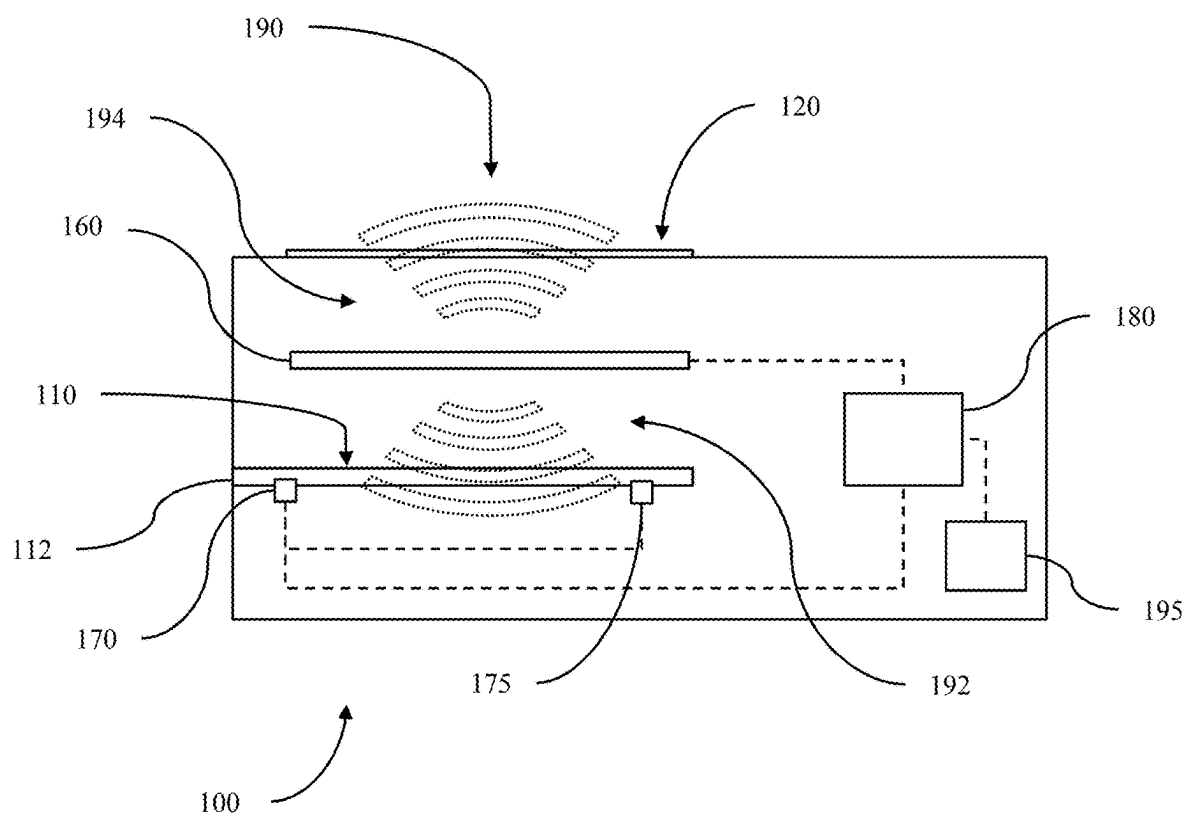


FIG. 2

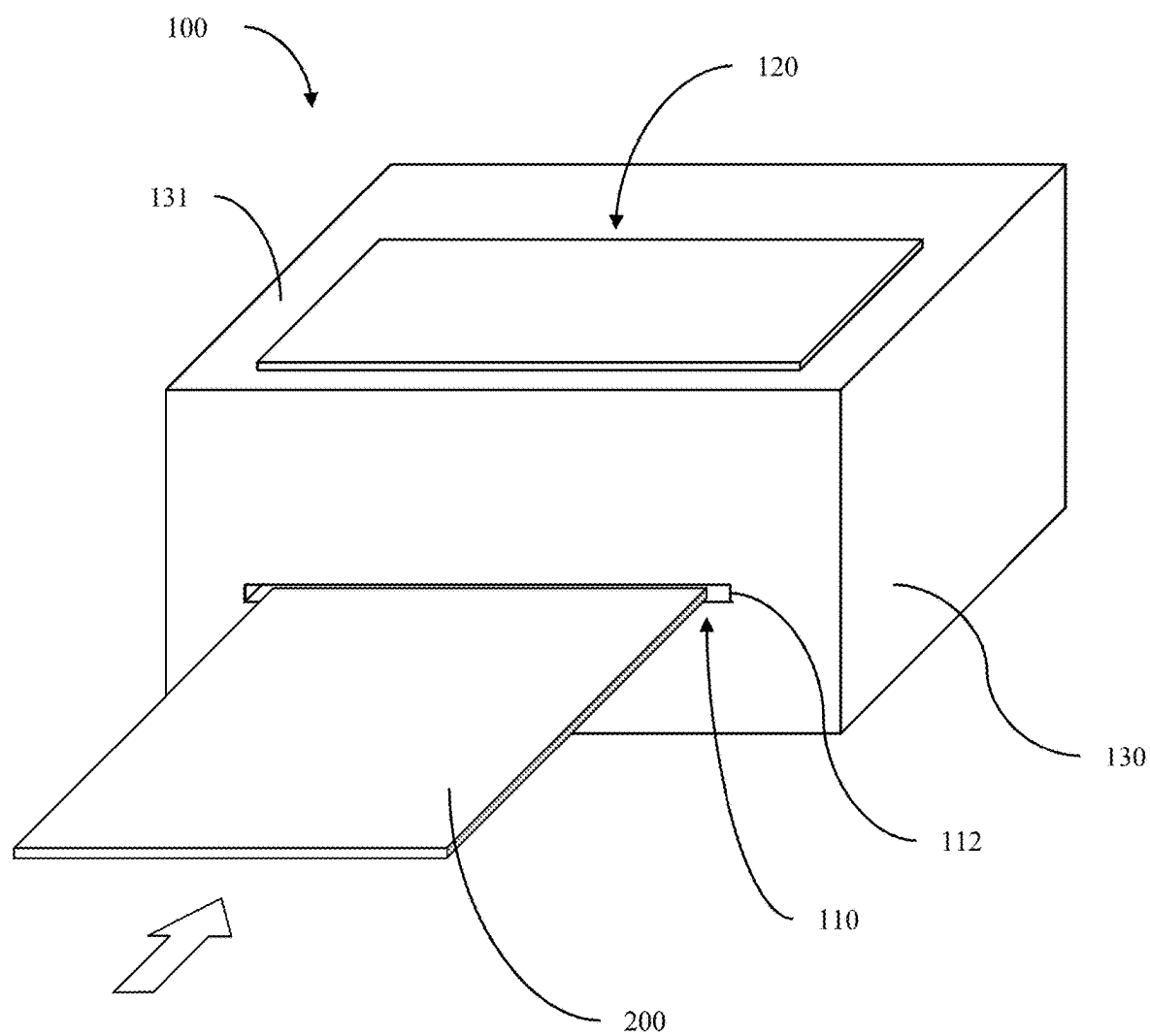


FIG. 3

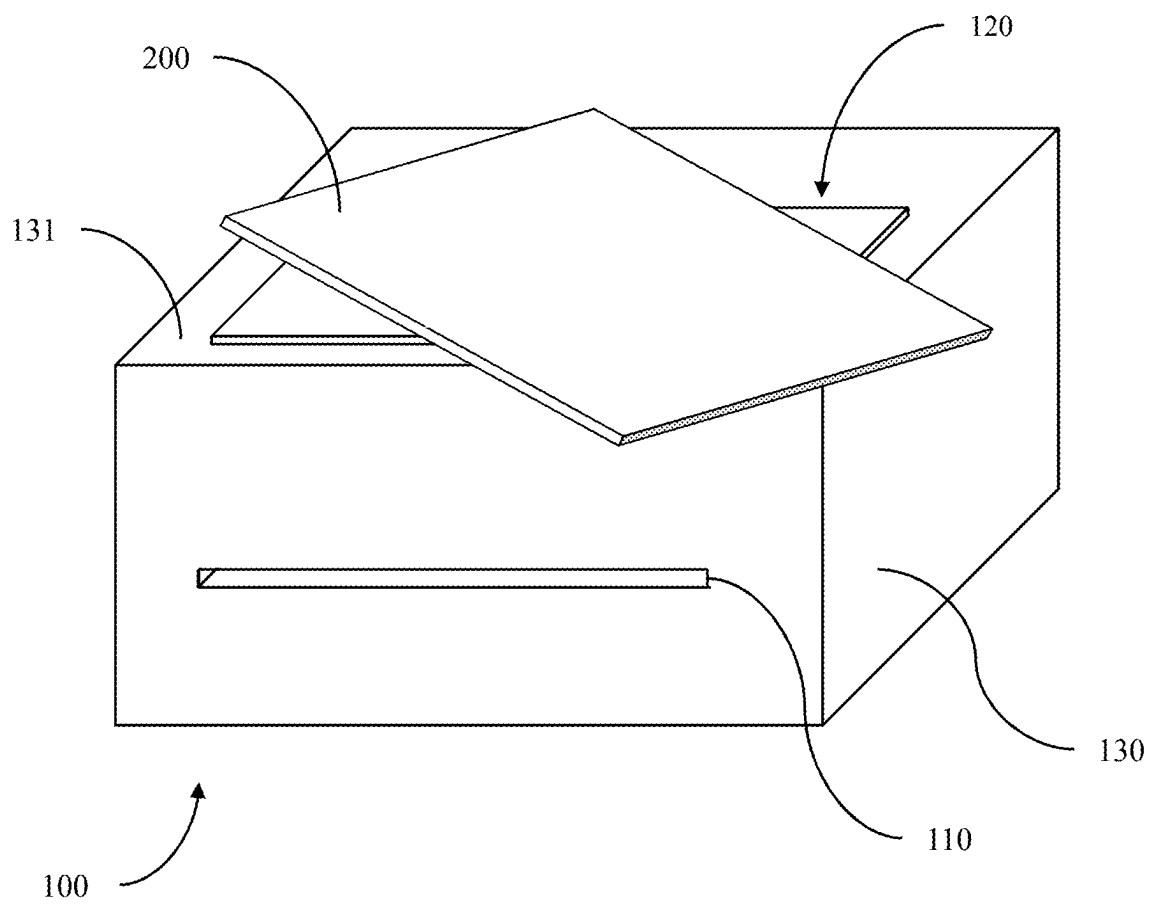


FIG. 4

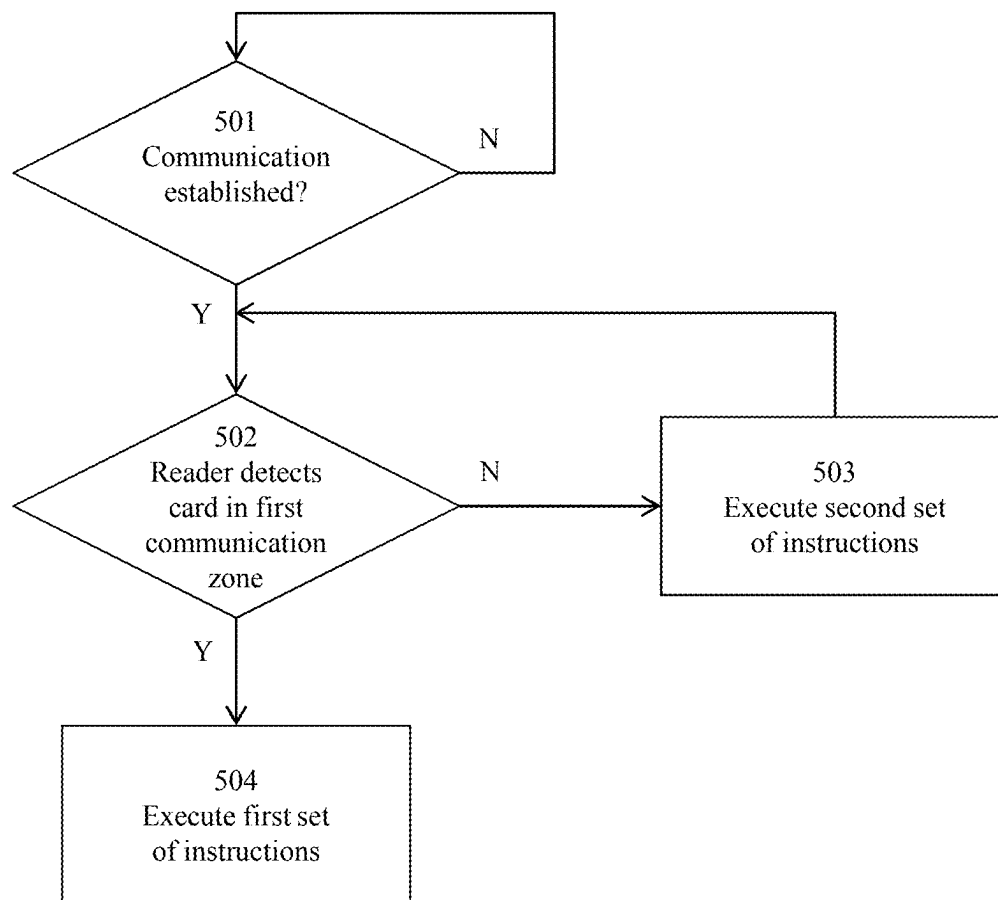


FIG. 5

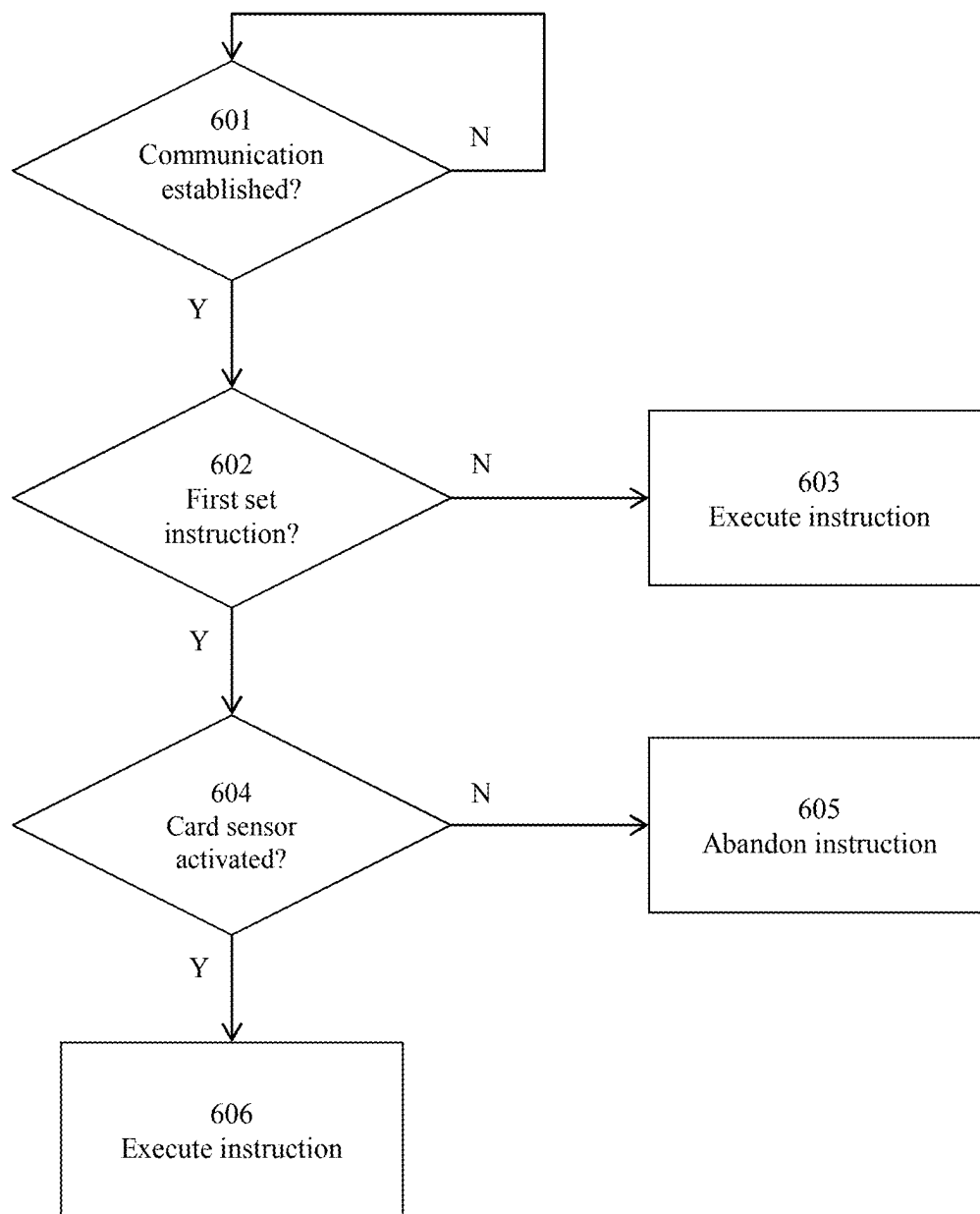


FIG. 6

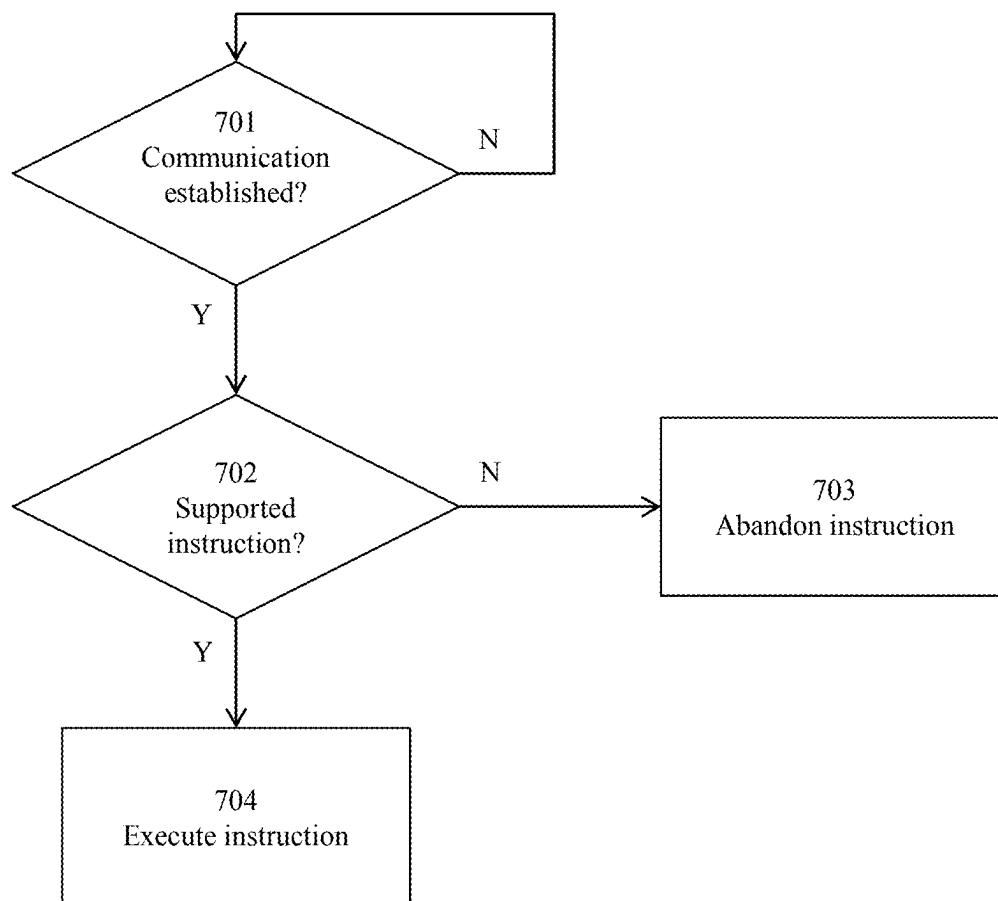


FIG. 7

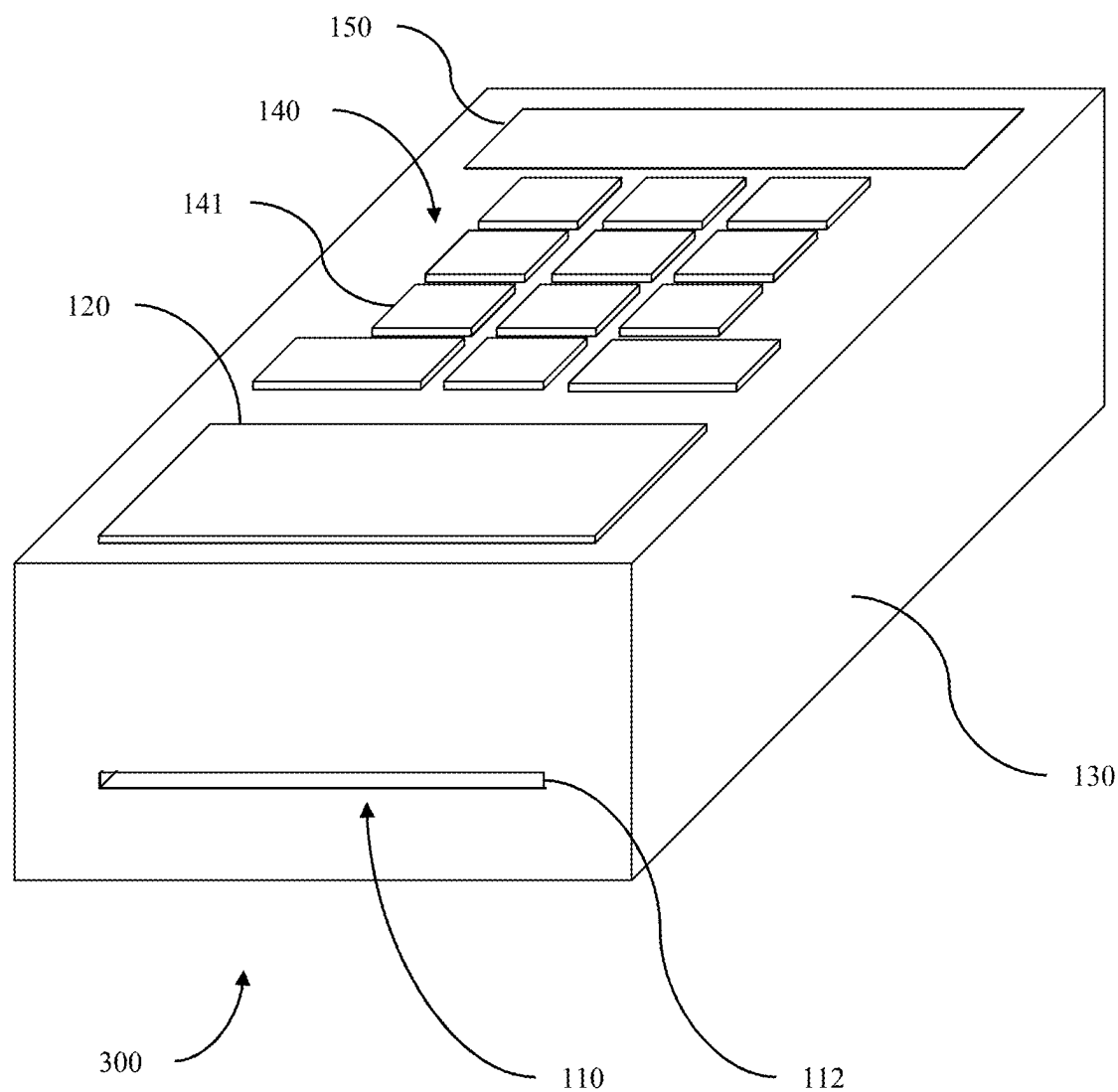


FIG. 8

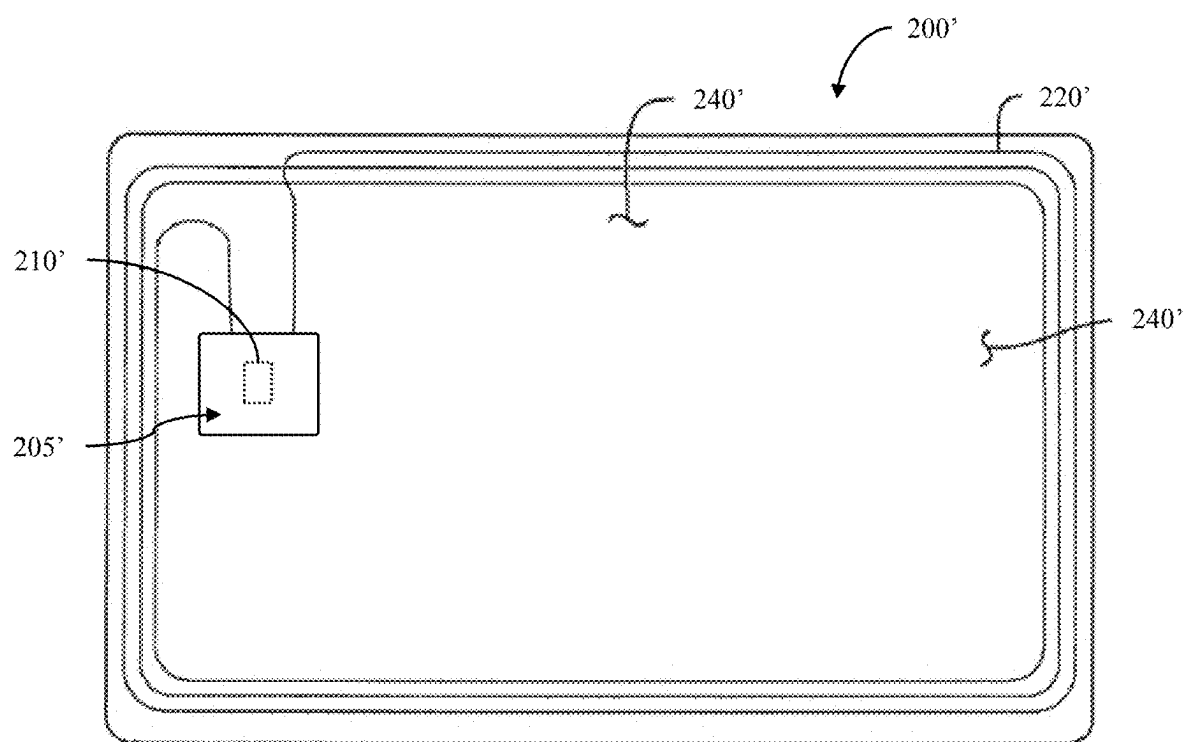


FIG. 9

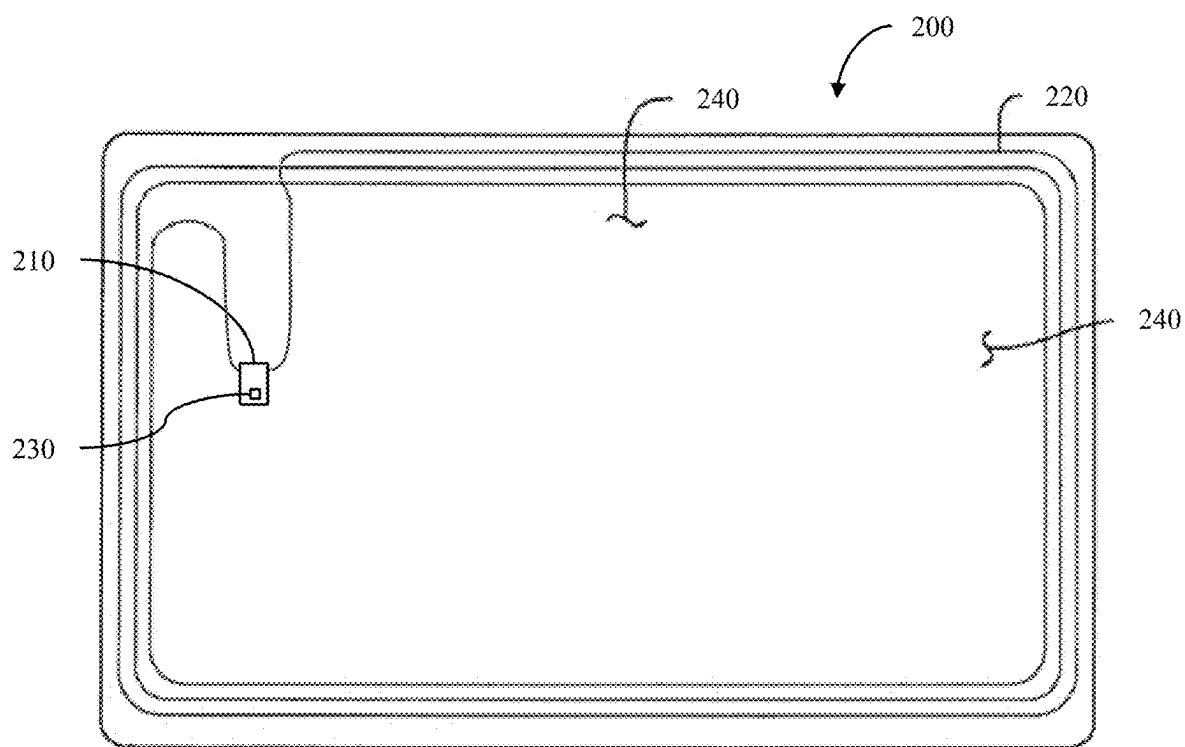


FIG. 10

CARD READER, SMART CARD AND METHOD FOR PROCESSING A TRANSACTION

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application is a continuation of U.S. patent application Ser. No. 18/110,420, filed on Feb. 16, 2023, which is a National Stage Entry of PCT/US2021/042321 filed on Jul. 20, 2021, and which claims the benefit of and priority of United Kingdom Patent Application No. 2012833.6, filed on Aug. 17, 2020. The entire disclosures of the above-mentioned applications are herein incorporated by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to a card reader, for example a point of sale device, a smart card, for example a payment smart card, and a method for processing a transaction.

BACKGROUND

[0003] Smart cards are cards that comprise an integrated circuit, allowing them to hold information and engage in communication with external systems. Contactless smart cards communicate with card readers using radio based communication whereas contact cards have electrical contacts to form an electrical contact with corresponding contacts of a card reader in order to communicate. Dual function smart cards (i.e. smart cards having both contact and contactless mechanisms) comprise components to allow them to operate using either interface. A dual-function smart card must also have a kernel and applications stored within it to enact contact transactions and contactless transactions.

[0004] Payment cards (such as credit and debit cards) are commonly dual function smart cards. This allows a cardholder (i.e. a user of the card) to interact with a payment terminal via either of the interfaces. Typically historically, the contactless interface is used for low value transactions and does not require any extra cardholder authentication. The contact interface is typically historically used for higher value transactions and requires cardholder authentication (such as provision of a Personal Identification Number (PIN)). The contact interface is also known to be used for verifying a biometric template, when used to authenticate the cardholder, and authorising changes security or financial details relating to the cards, such as for changing security information (e.g. the PIN and biometric template) or loading card balance. These transactions often require a longer interaction, either because multiple steps are taken as part of the transaction, or in that more data may be transferred from the terminal to the card or vice versa.

[0005] Normally the card allows a cardholder to perform operations on their bank account—for example transfer money, allow a remote bank account server to cause a balance for the account to be displayed at a terminal, change a password stored/authenticated at a remote account server. It is not common to allow changes to information stored on the card itself. The card can be thought of as a key that allows either off-card processes to take place (or not if the card is not present).

[0006] Dual-function smart cards and card readers are, by necessity, more expensive and more complex than their

single function counterparts as they must contain the necessary components to encompass both functions.

STATEMENTS OF INVENTION

[0007] One aspect of the invention provides a card reader for reading a smart card. The card reader comprises an antenna for making a wireless connection with a smart card, and a card detector, for determining a location of the smart card relative to the antenna. The card reader is configured to communicate with the smart card and to allow or provide a first wireless communication when the card is determined to be at a first location of the card reader and to provide a second different wireless communication when the card is determined to be at a second, different, location of the card reader.

[0008] Providing a different communication may comprise selecting one communication to provide to or from the card from two or more communications. It is often going to be the case that the different communications will have different functions or characteristics and may require different levels of security.

[0009] Smart cards are cards for controlling access to resources (for example financial, data or locations) which possess an integrated circuit. In particular, smart cards in the context of this specification primarily relate to smart cards for performing financial transactions—such as credit or debit cards.

[0010] The card reader may comprise a first communication zone. The first communication zone may comprise a card receiving and holding station, adapted to receive a card and hold it in place, possibly without a cardholder holding the card in place, and possibly without anyone touching the card. The first communication zone may comprise a pseudo-contact card-dipping station.

[0011] The card reader may comprise a second communication zone. The second communication zone may comprise a card contactless station.

[0012] The first and second locations of the card reader may comprise the first and second communication zones.

[0013] The card contactless station may be a region of a housing of the card reader where transactions may be carried out in that region through a cardholder “tapping” a card on the card reader in that location, or otherwise approaching the card contactless station with a card. The card contactless station may have an indication to a user that a card should be brought into the vicinity of the reader at that location. Transactions may be carried out without an actual physical tap; rather the card may be brought proximal to the card reader in the vicinity of the card contactless station, for example hovering at the contactless station. Of course, the user could touch the card contactless station with the card: it will still work.

[0014] The card detector may be configured to determine whether the smart card is within the first communication zone or the second communication zone, and to enable/disable certain functionality or operations that can be performed between the card and the card reader depending upon where the card is located.

[0015] The card reader (or a processor of the card reader or in communication with the card reader) may be looking to see if the card has started communicating via the antenna at the first or second communication zone. A card detector may be provided to enable it to be determined whether the card is in the first or second communication zone. For

example, if the card reader is expecting a card to be dipped into the first communication zone, the card detector at that zone may tell the card reader that the card is present, and that information may be used to trigger the pseudo-contact mode.

[0016] The first and second communication zones may be arranged on opposite sides of the antenna. The first communication zone may be underneath the antenna when the card reader is in use. The second communication zone may be above the antenna when the card reader is in use. The second communication zone may instead encompass any region (or even the entire region) around the antenna except a region defined by the first communication zone. The first and second communication zones may be arranged on the same side of the antenna.

[0017] The card detector may be configured to detect whether the smart card is in the first communication zone or not.

[0018] The card reader may determine that the smart card is in the second communication zone. It may do this by determining if the antenna can connect to the smart card and the card detector does not detect the smart card in the first communication zone.

[0019] The card detector may comprise a switch which generates a signal to a control processor when switched. The switch may also or instead activate a signal emitter of the card reader. The signal emitter is located in the first communication zone and is configured to provide a signal from which the smart card can determine that it is in the first communication zone. The signal emitter may be a light signal emitter, such as an LED, and may be configured to provide a specific wavelength of light. The signal emitter may be instead a magnetic field, for example a constant magnetic field generated by a small permanent magnet or an activated coil. The signal emitter may instead be an electrical contact, configured to make a connection with a corresponding electrical contact of the card.

[0020] If an electrical contact signal emitter is provided to enable the card to determine (itself determine) that it is in the first communication zone, that is not the same as a prior art contact card reader. The electrical contact signal emitter does not power the card or read/write card identity or security or financially related data to or from the card, and instead provides a logic state (0 or 1) to indicate to the card that the card is in the first communication zone (or not).

[0021] The switch is triggered by insertion of a card into the card receiving and holding station.

[0022] The switch could detect when a card is partly in the card holding and receiving station (sometimes considered a dipping zone), for example as it is beginning to be inserted, or it could detect whether the card is fully inserted into the card receiving and holding station, or it could do both. The card reader may make preparations to communicate in pseudo-contact mode when the card is detected as being partly in position but not actually make the communication with the card until the card confirms it supports the pseudo-contact mode. The card itself may determine that it is in the card holding and receiving station before to confirm it supports the pseudo-contact mode. The card reader may abandon the communication if it doesn't receive the confirmation that the card supports the pseudo-contact mode within a given timeout.

[0023] In one embodiment, the card will not start to communicate with the reader unless there is enough cover-

age or coupling with the antenna. When the signal is strong enough (before the card is fully slid home in the pseudo-contact slot for example) the card and card reader begin to communicate. However, only when the card has detected that it is fully inserted will it accept/authorise the operation of activities reserved for the pseudo-contact mode of communication.

[0024] The switch may consist of a mechanical switch (such as a micro switch or a tactile switch) or an optical switch. The switch may be provided at the aperture of a card-dipping slot in the card reader.

[0025] The card itself may have a sensor, for example a light sensor, to determine whether the card is in the dipping zone/card holding and receiving station.

[0026] Other sensor types (such as an electrical contact or a magnetic sensor) may be used. Any sensor that is operable to allow the card to detect that it is inserted in the dipping zone may be suitable.

[0027] The card reader (and/or the card itself) may be configured to provide communications of longer duration between the card reader and the smart card when the card is in the first communication zone (e.g. dipping zone) than when it is in the second communication zone (e.g. tapping zone).

[0028] Communications of longer duration provide for transfer of larger packages of data, and/or more complex series of human-machine interactions that can take longer to complete.

[0029] The data transfers in the first communication zone may last longer than 300 ms. Preferably the data transfers in the first communication zone last longer than 1 second.

[0030] The data transfers in the second communication zone may last less than 50 ms. Preferably the data transfers in the first communication zone last less than 300 ms.

[0031] Communications requiring longer durations may include one or more of at least: verifying a PIN, verifying a fingerprint, changing a PIN, updating data on the smart card, updating applications on the smart card, or updating security settings (for example changing keys).

[0032] Communications requiring larger data transfers may include one or more of at least: sending cardholder data (e.g. biometric data) to or from the smart card, updating card applications and sending or receiving other larger packages of data.

[0033] The card receiving and holding station may be configured to hold the card in a fixed position relative to the antenna, possibly without the user holding the card.

[0034] The card reader may have multiple antennae, but in some preferred embodiments the card reader may comprise no more than one antenna for communicating with a smart card. In other words, the card reader may comprise only a single antenna capable of communicating with a smart card, wherein the single antenna is operable to communicate with a smart card in two different modes.

[0035] Another aspect of the invention provides a card reader for reading a smart card. The card reader comprises an antenna for making a wireless connection with a smart card, a first communication zone, comprising a card receiving and holding station, and a second communication zone, comprising a card tapping or approaching station. The antenna is configured to communicate with a smart card provided at either the first communication zone or the second communication zone. The communication acts per-

mitted by the cardholder of the card via the card reader differ depending on which zone the card is provided at.

[0036] Another aspect of the invention provides a smart card for use with the card readers of the preceding aspects.

[0037] Another aspect of the invention provides a smart card. The smart card is configured to receive a signal from a card reader, determine which one of a number of different wireless communication positions relative to the card reader the card is positioned at, using the received signal, determine a wireless communication in dependence on the determined position or received signal, and provide or receive the wireless communication to or from the card reader.

[0038] This enables a smart card to permit certain functions/operations/interactions with the reader only when it has itself determined that it is in a specific position relative to the card reader. This may be achieved by the card having a logic processor capable of making such a determination and authorizing particular functionality when it determines that it is in the right position.

[0039] The smart card of either of the preceding aspects may comprise a memory containing a first set of instructions and a second set of instructions, wherein the first set of instructions provides a first communication and the second set of instructions provides a second communication. The first set of instructions may comprise the second set (that is to say all instructions may be enabled/authorised at the first location and only a part of the instructions at the second location relative to the card reader).

[0040] The smart card may comprise a sensor for receiving the signal from the card reader.

[0041] The sensor may comprise a light sensor, or a magnetic sensor, or an electrical contact.

[0042] Another aspect of the invention provides a method of processing a transaction. The method comprises determining the location of a smart card relative to an antenna of a card reader; and providing a first wireless communication if the card is determined to be at a first location, or providing a second wireless communication if the card is determined to be at a second location.

[0043] In many embodiments, it is the card that determines which of the first or second wireless communications to make.

[0044] It will be appreciated that the first and second communicators may enable different card reader interactions.

[0045] Another aspect of the invention provides a data processing apparatus/device/system comprising a processor configured to perform the method of the preceding aspect.

[0046] Another aspect of the invention provides a computer program comprising instructions which, when the program is executed by a computer, cause the computer to carry out the method of the above aspect.

[0047] Another aspect of the invention provides a non-transitory computer-readable storage medium comprising instructions which, when executed by a computer, cause the computer to carry out the method of the above aspect.

[0048] Another aspect of the invention provides a payment card comprising a wireless emitter and receiver (transceiver) and a processor, the processor being adapted to receive information relating to the position of the card relative to a card reader when a data transfer between the reader and the card is to be performed, and adapted to enable or disable the performance of specific actions available to a cardholder depending upon the location of the card.

[0049] The processor on the card may be adapted to receive a signal from, or derived from, a card-position sensor associated with a card reader, and to use that signal to enable the card to perform particular interactions with the card reader.

[0050] The card itself may have a sensor adapted to sense when the card is in a card reader slot and which is in communication with the processor, the processor being configured to enable at least one specific action available to the cardholder only when it receives a signal from the sensor indicating that the card is in the card reader slot.

[0051] Another aspect of the invention provides a method of reducing the costs of manufacturing payment cards having a contactless communication capability and a contact communication capability. The method comprises not including a true contact infrastructure in the card, and instead having a processor or chip which is configured to determine whether the card is in a card reading slot of a card reader and use wireless communications with the card reader to perform restricted card reader interactions permitted only when the card is in the slot, so as to provide pseudo-contact capability.

[0052] Another aspect of the invention provides a method of reducing the costs of manufacturing payment card readers having a contactless communication capability and a contact communication capability. The method comprises not including a true contact infrastructure in the card reader, and instead having a processor or chip which is configured to determine whether a payment card is in a card reading slot of the card reader and use wireless communications with the card to perform restricted card reader interactions permitted only when the card is in the slot, so as to provide pseudo-contact capability.

BRIEF DESCRIPTION OF THE DRAWINGS

[0053] A description of the invention will now be described, by way of example only, with reference to the accompanying drawings, in which:

[0054] FIG. 1 shows an oblique schematic view of a card reader;

[0055] FIG. 2 shows a simplified sectional view of the card reader of FIG. 1;

[0056] FIG. 3 shows an oblique schematic view of a smart card being used in a first manner with the card reader of FIG. 1;

[0057] FIG. 4 shows an oblique schematic view of a smart card being used in a second manner with the card reader of FIG. 1;

[0058] FIG. 5 shows a process flow for using a smart card with a card reader;

[0059] FIG. 6 shows a subsequent process flow for using a smart card according to examples of the invention with a card reader according to examples of the invention following the steps of FIG. 5;

[0060] FIG. 7 shows another subsequent process flow for using a legacy smart card with a card reader according to examples of the invention following the steps of FIG. 5;

[0061] FIG. 8 shows an oblique schematic view of a further card reader;

[0062] FIG. 9 shows a sectional schematic view of a legacy contactless/dual interface smart card; and

[0063] FIG. 10 shows a sectional schematic view of a contactless smart card according to examples of the invention.

DETAILED DESCRIPTION

[0064] FIG. 1 shows a card reader 100 (otherwise known as a “payment terminal” or “point of sale device”) for communication with a smart card. The card reader 100 is shown in an in-use orientation, in which a front of the device is facing the viewer and the top of the device is orientated towards the top of the page.

[0065] The card reader 100 comprises a housing 130 and two communication zones. A first communication zone 110 is constructed to superficially resemble card reading zones of conventional contact card readers. The first communication zone comprises a card receiving and holding station in the form of a slot, having an opening 112 in the housing 130 through which a card may be at least partially inserted. The opening 112 is formed in the front surface of the device. A second communication zone 120 is provided externally to the housing. The second communication zone 120 is situated on and proximal to a surface 131 of the housing 130. The surface could be any surface, but in this example the surface is the top surface 131 of the housing. Using the top surface 131 as the second communication zone further enforces the visual similarity and cardholder perception of functionality between the card reader 100 and card readers known in the art. The communication zones 110, 120 are regions in which a smart card may be placed in order to communicate with the card reader 100. The content of the communication is dependent upon which communication zone the smart card is placed in, as described in more detail below. Depending upon which communication zone is used the smart card permits some operations and not others; some interactions with the card are reserved for taking place only when the card is in a selected specified zone.

[0066] A simplified cross section of the card reader 100 is shown in FIG. 2. The card reader 100 comprises a single card communicator 160. The communicator is an antenna 160 that functions as a wireless emitter/receiver/transceiver. This is different to conventional, dual interface, card readers which possess a pair of card communicators, one for contact transactions (e.g. chip and pin) with electrical contacts and one for contactless transactions with an antenna. The antenna 160 is similar in geometry and size and thickness to a card, and is arranged so as to provide a communication field 190 that encompasses both the first and second communication zones 110, 120, so that a card can be read and/or written to wirelessly/in a contactless manner whichever zone it is in. The first and second communication zones are arranged such that they are on different sides of the antenna. The first communication zone 110 is situated below the antenna and the second communication zone 120 is situated above the antenna 160. The field 190 has a first region 192 below the antenna and a second region 194 above the antenna.

[0067] A card detector 170 is provided within the first communication zone 110. The card detector in this example consists of a switch 170. In this example the switch 170 is a mechanical switch, but any switch that may be triggered by insertion of a card into the card reader may be suitable, for example, an optical switch. The switch 170 is triggered when a card is inserted into the slot. The switch 170 is in communication with a processor 180, to which it provides a signal indicating the presence (or lack of presence) of a card in the slot. This switch is also in communication with a signal emitter 175. The signal emitter 175 is configured to provide a signal to the card 200 in order that the card 200 can

determine that it is within the first communication zone. The signal emitter is provided as an LED configured to emit a light for detection by the card 200. Other light sources may be used. The light is provided over a specific range of wavelengths (for example ultra violet). In other examples other signal emitters may be used, for example a permanent magnet or a magnetic coil generating a constant magnetic field in the card reader detected by a magnetic field sensor in the card, or an electrical contact, configured to make an electrical connection with a corresponding electrical contact on the card 200. The electrical contact is not as complex as the contacts used in contact smart cards, which are used for transferring data, but rather consists in this example of a single pin contact to simply provide a signal indicative of insertion of the card into the slot (or of the presence of the card in the slot) to the card 200. The processor 180 is also in communication with the antenna 160. The processor 180 is configured to execute instructions in order to provide the antenna with data for communication with the smart card and similarly to receive data from the smart card. In some examples the card detector 170 may not be in direct communication or any communication with the processor 180. Rather the card detector 170 simply triggers the signal emitter 175 by closing a circuit, without any processing requirements. In still further examples a card detector 170 is not provided and instead the signal emitter 175 exists in an ‘always on’ state provided the card reader is powered.

[0068] A software-processing kernel of the processor 180 of the card reader is configured to provide operation in two modes. A first mode is used if it is determined that a smart card is in the first communication zone. If a card is determined to be within the communication field 190 and the card detector 170 detects the presence of a card then a determination that the card is in the first communication zone 110 is made. If a card is in the communication field but the card detector 170 does not detect a card, then it is determined that the smart card is in the second communication zone 120. The second mode is used if a card is determined to be in the second communication zone.

[0069] The card reader 100 provides a single antenna through which a connection can be established with a smart card whilst still appearing to the cardholder to provide a dual-function card. Conventional card readers require distinct different infrastructure for each function. The card reader 100 is therefore simpler and cheaper to construct than conventional card readers. Also, prior art cards require contact points for contact reading/writing, and also wireless transceivers. A card with only a wireless transceiver and no contact structure is less expensive to make. Not having a complex contact structure also allows the card to have a slimmer profile and be of lighter construction, and therefore reducing storage requirements in a cardholder’s wallet or purse. Another useful feature is that the pseudo-contact card reader slot 112 can accommodate cards of different thicknesses, which is not the case with true contact card reader slots. This gives the opportunity to card manufacturers to provide thinner and lighter cards.

[0070] Only a single kernel is also required to be hosted in the processor 180. The single kernel is operable to define the different use cases and modes of operation. This further simplifies the operation of the processor. Newer cryptographic evolutions on contactless smart card will soon also make contactless transactions more secure than contact transactions. Only using the contactless interface therefore

improves the security of transactions. The card reader **100** emulates the function of conventional card readers so as to provide cardholder with a process of carrying out a transaction to which they are already accustomed. This maintains cardholder confidence in the security of transactions and also negates the need for cardholder training.

[0071] If we consider legacy support for existing cards, the card reader will support contactless interactions with existing contactless cards (contactless only or dual interface), but will not support true contact interactions with legacy cards (contact only or dual interface). In some embodiments, the card may still have a contact interface to interact with existing true contact interface card readers, whilst also having the capacity to interact wirelessly with pseudo-contact interface card reader.

[0072] The card reader also comprises a connector **195** that provides a connection to a network. The connector **195** is in the form of a Wi-Fi® antenna, but other connectors may be used, such as over other wireless connections (e.g. Bluetooth®) or through a wired connection. Information may be provided from the network (and/or a memory or a computing device also connected to the network) to the card reader **100**. The card reader **100** is configured to operate online or offline (i.e. connected or disconnected to the network) in order to enact transactions.

[0073] FIG. 3 shows a smart card **200** being inserted into the slot associated with the first communication zone **110**. The cardholder interaction with the card reader **100** is configured to be similar or even the same as the “dipping” interaction with a conventional chip and pin card reader, despite the differing mechanisms. This may include requiring further authentication before subsequent transaction/acts are allowed. The further authentication might be the input of a Personal Identification Number, a thumb print scanner or other biometric data or any other cardholder authentication means. As the card is inserted into the slot it enters the first communication zone **110** and the communication field **190**, and communicates with the processor **180** via the wireless connection, using the first region **192** of the field **190** that is below (to one side) of the antenna **160**.

[0074] FIG. 4 shows a smart card being placed in the second communication zone **120**. The cardholder interaction with the card reader is configured to be similar or even the same as the “tap” interaction with a conventional contactless card reader despite the differing mechanisms. The card **200** communicates with the reader **100** using the second region **194** of the field **190** that is above (to the other side) of the antenna **160**.

[0075] FIG. 5 shows a process flow for a method of enacting a transaction using a smart card and a card reader.

[0076] At a first step **501** a determination is made as to whether a card has established a communication with a card reader. If not then the process resets. If yes then the process moves to a second step **502** in order to determine whether the card is in a first communication zone or a second communication zone. If the card is determined to be in the first communication zone, then the process moves to a third step **503**. If the card is not determined to be in the first communication zone, then the card is instead determined to be in the second communication zone and the process moves to a fourth step **504**. The third and fourth steps **503**, **504** comprise executing a first or second set of instructions respectively. The first set of instructions provides the steps for enacting a communication between the card and the card

reader when the card is in the first communication zone. The second set of instructions provides the steps for enacting a communication between the card and the card reader when the card is in the second communication zone. The first and second sets of instructions are comprised within a single kernel hosted on the card reader and/or within a single kernel hosted on the card. When executing the second set of instructions the card may be determined to be in the first communication zone then the process moves to step **503**.

[0077] The second step **502** comprises the card (itself) detecting a signal provided from the card reader. The signal is only detectable when the card is physically inserted into the card reader, requiring alignment of corresponding components of the card and card reader or close proximity (wherein close proximity is considered to be between contact and 5 mm). This prevents the card from determining that it is inserted in the dipping zone of a card reader, as described in the invention, and performing restricted card reader interactions when in fact the card is still in the cardholder wallet or purse.

[0078] The card itself has enough processing capability on it to receive a signal indicative of the card being in the correct place in an allowed card reader and permitting on itself appropriate processing to enable restricted card reader interactions to take place, or to send an enable signal to the card reader enabling the card reader to perform restricted card reader interactions.

[0079] The steps flowing from the activation of the first or second set of instructions are shown in FIGS. 6 and 7 respectively.

[0080] In FIG. 6 the card has established the communication with the card reader has been determined by the card reader to be in the first communication zone and therefore the first set of instructions is executed. As the card is in the first communication zone it is held stationary relative to the antenna, thereby providing a more stable connection than if the card was not held stationary. Typically, cardholders push the card into the slot **112** and let go of it there, leaving it held in the reader **100**. This allows for communications that are longer in duration and therefore can comprise larger amounts of data transfer. The connection established in step **501** allows a communication between the card and the card reader at step **601**. The communication defines the nature of the transaction to be carried out. Transactions to be carried out which may require longer durations in order to process or transfer larger amounts of data transfer or to have a series of human-machine interactions are likely to be more sensitive in nature, for example updating details on the card or changing a PIN. The transactions may take longer than 300 ms, or even longer than 1 second.

[0081] The instruction is received in a first step **601**. In a second step **602** a determination is made as to whether the instruction is of the first set of instructions. If the instruction is not of the first set, then it is automatically determined that it is therefore of the second set of instructions and can therefore be executed in a third step **603**. If the instructions are indeed of the first set, then the process moves instead to a fourth step **604**. In the fourth step **604** a determination is made as to whether the card sensor is activated. If the card sensor is not activated then the instruction is abandoned in a fifth step **605**. Abandoning the instruction provides a security measure against unwarranted transactions. If the card sensor is activated then the process instead moves to a sixth step **606** of executing the instructions. As the card

sensor is activated then the security measures of confirming the card is in the reader is met and more sensitive transactions can therefore be carried out.

[0082] In FIG. 7 a legacy card is instead used with the card reader. As with FIG. 6, the first step 701 comprises the card receiving instructions. A determination is then made at a second step 702 as to whether the instruction is a supported instruction. The legacy card will still be able to carry out legacy contactless functions, such as low value payments, and so some functions may be available. If the instruction is not supported then the process moves to a third step 703 and the instruction is abandoned. If the instruction is supported then the process instead moves to a fourth step 704 and the instruction is executed.

[0083] In both of the processes of FIGS. 6 and 7 a confirmation that the transaction was carried out successfully may also be provided to the cardholder, to a financial services provider, or both. Similarly, a failed transaction notification may also be provided should the transaction be unsuccessful.

[0084] FIG. 8 shows a further example of a card reader 300. The card reader 300 is similar in construction to the previous card reader 100, having a housing 130, a first communication zone 110, and a second communication zone 120. The card reader 300 further comprises a keypad 140. The keypad 140 allows a cardholder or vendor to input data. The input data may be in response to queries or requests displayed on a display 150 of the card reader 300, such as a request for a PIN. The keypad 140 comprises physical keys 141 in this example. Other examples of card readers 300 may instead use touch screens, computer peripherals, or any other suitable keypad by which a cardholder may interact with the card reader 300.

[0085] FIG. 9 shows a prior art legacy smart card 200' having a chip 210', a wireless antenna 220' for wireless communication with a card reader, and a module 205' supporting the chip 210' with electrical contacts for contact communication between a card reader with electrical contacts and the card. The card has a surface 202' that is opaque.

[0086] If the card 200' is used in the card reader of FIG. 1-4 or 8, it can communicate wirelessly via its antenna 220'. Whether it is in the first or second communication zone the card communicates wirelessly. No contact communication takes place via the contact 205' because the card reader has no matching contacts.

[0087] FIG. 10 shows a smart card 200 in accordance with the invention. The smart card is in the main of conventional construction for contactless smart cards and is configured for use with the example card readers described above at least.

[0088] The smart card comprises a chip 210 which is configured to store information relating to the associated bank account and the cardholder of the smart card. The chip 210 also stores instructions for communicating with a card reader in either a first mode or a second mode (i.e. with a first communication or a second communication). The first mode provides operation of the card 200 in a pseudo-contact scenario and the second mode provides operation of the card 200 in a contactless scenario. A main antenna 220 is provided in the card in order to provide a means by which the card can be powered by and communicate with the card reader. The antenna 220 is connected to the chip 210 and is configured to transmit and receive data in dependence upon the operation of the card.

[0089] The chip 210 is configured to determine whether it is in the first communication zone 110 or the second communication zone 120. As shown in FIG. 10, the chip comprises a sensor 230 that may be configured to receive a signal directly from the signal emitter 175 when the card is inserted into the slot 110.

[0090] The sensor 230 in one example is a light sensor, configured to sense a light emitted by the card reader within the slot. The light sensor 230 may be in the form of an electronic circuit with a photoelectric diode attuned to a specific frequency—such as ultraviolet (UV) or infrared (IR)—of which the card reader is configured to provide (with an LED for example). A portion of the surface 240 of the card covering the light sensor is translucent or transparent in order to allow the ingress of light to the sensor.

[0091] In another example the sensor 230 comprises a magnetic sensor, configured to sense a magnetic field generated by the card reader within the slot. The magnetic sensor may be in the form of an electronic circuit with a Hall Effect transistor which switch when close enough of a magnetic field—which the card reader is configured to provide (with a permanent magnet or an activated coil for examples).

[0092] In another example the sensor 230 comprises an electrical contact which corresponds to an electrical contact of the card reader within the slot. When the contacts are connected the card makes the determination that it is within the first communication zone. The electrical contact in the card is less complex than prior art contact—read contacts in cards. It simply needs to enable the card to know that it is in the right position in a card reader, not transfer data.

[0093] Using an electrical contact, a magnetic sensor or a light sensor (or any other sensor that requires close proximity) as the sensor provides an additional layer of security as restricted card reader interactions cannot be activated when situated in a cardholder's pocket, wallet, purse or bag. This, therefore, prevents theft of data or funds through emulating a pseudo-contact mode as it is not possible to activate the card without it being removed from and being inserted into the card reader configured with a trigger for the card sensor (e.g. a corresponding electrical contact, a magnet or an LED).

[0094] The card receives power from the electromagnetic field of the card reader. As such the chip and the sensor of the card are only powered when the card is in proximity to the card reader. This type of proximity is enough to power the card in the cardholder wallet or purse with a card reader but not to activate the restricted card reader interactions locked by the card sensor.

[0095] There can be also a value in enabling some—but perhaps not all—functionality of the card reader when the card is inserted in the dipping zone even if the card has no sensor to detect it is inserted in the dipping zone.

[0096] There can be 3 use cases depending of the activation of the reader switch and the card sensor:

[0097] Both the reader and the card detect that the card is in the dipping zone—the card according to the invention fully supports the pseudo-contact mode: long transfer operations are enabled (for example reading encrypted card files, such as a picture to identify the cardholder, fingerprints to be checked on the reader connected to a server) and restricted functionality are enabled (for example verify PIN/biometrics in the card, data and/or security changes) are supported.

Legacy cards do not support the pseudo-contact mode of operation and so those cards will not recognize related instructions.

[0098] The reader detects the card is in the dipping zone but the card does not—a card according to the invention supports partially (but not completely) the pseudo-contact mode: for example long transfer operations are supported, but restricted or reserved functionality are not. Legacy cards do not support pseudo-contact mode operations.

[0099] The reader communicates with the card wirelessly but it is not detected in the dipping zone—in this scenario the pseudo-contact mode is not supported.

[0100] It is also worth discussing the cardholder verification that may take place (for example Verify PIN or verify/match biometrics on the card). In a pseudo-contact transaction, this verification allows the transaction to proceed, and other restricted operations are also allowed which are controlled by the cardholder (Change PIN) or by the card issuer (Change Balance). In a contactless transaction, the Verify PIN command is disabled. It is not disabled specifically to prevent high value transactions, but rather to prevent an attacker from trying to guess the PIN of the card while it is still in the cardholder wallet or purse. If an attacker were to succeed in guessing the PIN whilst the card is in a wallet or purse, they could then read other card data useful in combination with the PIN, or even attempt to steal the card. If they don't succeed in guessing the PIN the process will lock the card and the cardholder can't use their card anymore (denial of services). Due to the short length of the PIN (4 digits), this kind of in-wallet/purse attack is considered as plausible even if it is essentially academic. The present invention disables some functionality if the operations are not executed in the proper environment. It is not securing the operations as such, more not letting them run at all. For instance, in some embodiments the invention blocks the Verify PIN command, so it blocks the transaction before the PIN has been entered. This means that the cardholder does not find that their card has been blocked because a hacker has incorrectly guessed the PIN too many times whilst it was still in their purse or wallet—the hacker cannot run the verify PIN command at all because the card will not let it run because it does not see the card reader signal at the card's sensor.

[0101] Various modifications can be made to the examples described above without departing from the scope of the appended claims. Features of the examples and embodiments may be exchanged, combined, omitted or adapted. The teaching of the specification should be taken as a whole with no limitation placed on scope of the appended claims by reference to the included description and drawings.

1. A smart card, comprising:
 - a card sensor configured to receive a signal from a card reader device;
 - a processor; and
 - at least one memory, comprising a plurality of program instructions which, when executed by the processor, causes the processor to:
 - determine whether the smart card is disposed within a first communication zone or a second communication zone of the card reader device based on the received signal; and
 - enable a wireless communication with the card reader device using a first mode of operation or a second mode of operation based on the determination.

2. The smart card of claim 1, further comprising an antenna configured to perform the wireless communication with the card reader device.

3. The smart card of claim 1, wherein the plurality of program instructions which, when executed by the processor, causes the processor to determine whether the smart card is disposed within the first communication zone or the second communication zone further causes the processor to:
 - determine the smart card is disposed within the first communication zone if the card sensor receives the signal; and
 - determine the smart card is disposed within the second communication zone if the card sensor does not receive the signal.

4. The smart card of claim 3, wherein the plurality of program instructions which, when executed by the processor, cause the processor to enable the wireless communication with the card reader device further causes the processor to:
 - enable the wireless communication with the card reader device using the first mode of operation based on a determination that the smart card is disposed within the first communication zone; and
 - permit one or more restricted card reader interactions via the wireless communication while using the first mode of operation.

5. The smart card of claim 3, wherein the plurality of program instructions which, when executed by the processor, causes the processor to enable the wireless communication with the card reader device further causes the processor to:
 - enable the wireless communication with the card reader device using the second mode of operation based on a determination that the smart card is disposed within the second communication zone; and
 - prevent one or more restricted card reader interactions via the wireless communication while using the second mode of operation.

6. The smart card of claim 1, wherein the plurality of program instructions which, when executed by the processor, causes the processor to determine whether the smart card is disposed within the first communication zone or the second communication zone further causes the processor to:
 - receive an instruction from the card reader device; and
 - determine whether the received instruction corresponds to a first instruction or a second instruction, wherein the first instruction corresponds to the first mode of operation and the smart card being disposed within the first communication zone, and wherein the second instruction corresponds to the second mode of operation and the smart card being disposed within the second communication zone.

7. The smart card of claim 6, wherein the plurality of program instructions which, when executed by the processor, causes the processor to enable the wireless communication with the card reader device further causes the processor to:
 - determine that the received instruction corresponds to the second instruction; and
 - perform the second instruction, comprising prevent one or more restricted card reader interactions via the wireless communication.

8. The smart card of claim 6, wherein the plurality of program instructions which, when executed by the processor, causes the processor to enable the wireless communication with the card reader device further causes the processor to:
 - enable the wireless communication with the card reader device using the second mode of operation based on a determination that the smart card is disposed within the second communication zone; and
 - prevent one or more restricted card reader interactions via the wireless communication.

sor, causes the processor to enable the wireless communication with the card reader device further causes the processor to:

- determine that the received instruction corresponds to the first instruction;
- perform the first instruction if the smart card is determined to be disposed within the first communication zone based on the received signal, comprising permit one or more restricted card reader interactions via the wireless communication.

9. The smart card of claim 6, wherein the plurality of program instructions which, when executed by the processor, causes the processor to enable the wireless communication with the card reader device further causes the processor to:

- determine that the received instruction corresponds to the first instruction; and
- abandon the first instruction if the smart card is determined to be disposed within the second communication zone based on the received signal.

10. A card reader device, comprising:

- a processor; and
- at least one memory, comprising a plurality of program instructions which, when executed by the processor, causes the processor to:
 - determine whether a smart card is disposed within a first communication zone or a second communication zone of the card reader device;
 - enable a first wireless communication with the smart card using a first mode of operation based on a determination that the smart card is disposed within the first communication zone; and
 - enable a second wireless communication with the smart card using a second mode of operation based on a determination that the smart card is disposed within the second communication zone.

11. The card reader device of claim 10, further comprising an antenna configured to perform the first wireless communication or the second wireless communication with the smart card.

12. The card reader device of claim 10, wherein the first communication zone comprises a dipping slot, wherein the smart card is disposed within the first communication zone when the smart card is at least partially inserted into the dipping slot.

13. The card reader device of claim 12, wherein the first communication zone further comprises a card detector, wherein the card detector is configured to generate a detection signal when the smart card is at least partially inserted into the dipping slot.

14. The card reader device of claim 13, further comprising a signal emitter, wherein the signal emitter is configured to generate a signal for the smart card based on the detection signal.

15. The card reader device of claim 14, wherein:

- the card detector comprises a switch;
- the signal emitter comprises a light source, a magnet, a magnetic coil, or an electrical contact; or
- combinations thereof.

16. The card reader device of claim 10, wherein the second communication zone comprises an area that is situated on and proximate to one or more designated surfaces of the card reader device.

17. The card reader device of claim 10, wherein the plurality of program instructions which, when executed by the processor, causes the processor to enable the first wireless communication with the smart card using the first mode of operation further cause the processor to permit one or more restricted interactions with the smart card via the first wireless communication while using the first mode of operation.

18. The card reader device of claim 10, wherein the plurality of program instructions which, when executed by the processor, causes the processor to enable the first wireless communication with the smart card using the first mode of operation further cause the processor to prevent one or more restricted interactions with the smart card via the second wireless communication while using the second mode of operation.

19. A system, comprising:

a card reader device, comprising:

a first processor;

at least a first memory, comprising a plurality of program instructions which, when executed by the first processor, causes the first processor to:

- determine whether a smart card is disposed within a first communication zone or a second communication zone of the card reader device;
- enable a first wireless communication with the smart card using a first mode of operation based on a determination that the smart card is disposed within the first communication zone; and
- enable a second wireless communication with the smart card using a second mode of operation based on a determination that the smart card is disposed within the second communication zone; and

a signal emitter, wherein the signal emitter is configured to generate a signal when the smart card is disposed within the first communication zone; and

the smart card, comprising:

- a card sensor configured to receive the signal from the signal emitter;
- a second processor; and
- at least a second memory, comprising a plurality of program instructions which, when executed by the second processor, causes the second processor to:
 - determine whether the smart card is disposed within the first communication zone or the second communication zone based on the received signal; and
 - enable the first wireless communication or the second wireless communication with the card reader device based on the determination.

20. The system of claim 19, wherein:

the card reader device further comprises a first antenna configured to perform the first wireless communication or the second wireless communication with the smart card; and

the smart card further comprises a second antenna configured to perform the first wireless communication or the second wireless communication with the card reader device.

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